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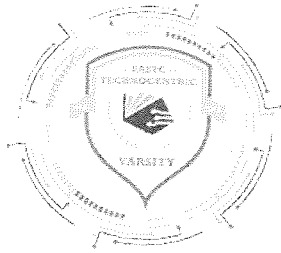
### **LEARNER FILE DIVIDER**

| CONTENT              | CHECKED |
|----------------------|---------|
| LEARNER WORKBOOK     |         |
| SUMMATIVE ASSESSMENT |         |

### **KNOWLEDGE MODULES**

| KM ID   | TITLE                                | Credits |
|---------|--------------------------------------|---------|
| KM – 08 | Low Voltage Protection, NQF Level 4. | 5       |





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30  
80

## LOW VOLTAGE PROTECTION

### SUMMATIVE ASSESMENT

|                                     |                                                  |
|-------------------------------------|--------------------------------------------------|
| <b>Module #</b>                     | 671101000-KM-08:                                 |
| <b>NQF Level</b>                    | level 4,                                         |
| <b>Notional hours</b>               | 50                                               |
| <b>Credit(s)</b>                    | Cr 5                                             |
| <b>Occupation or Specialisation</b> | 671101000: Occupational Certificate: Electrician |
| <b>Associated Occupation</b>        | 671101: Electrician                              |

| Details         | Please Complete this Section |
|-----------------|------------------------------|
| Name            | Xolani Collen Malinga        |
| Contact Address | 32467 Mandela Str G42B       |
| Telephone (H)   |                              |
| Telephone (W)   | 0747732317                   |
| Facsimile       | N/A                          |
| Cellular        | 0747732317                   |
| E-mail          | collenx14@gmail.com          |

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## **Welcome to your Summative Assessment!**

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

*Summative*

## LEARNER ASSESSMENT PLAN

| Unit<br>code | TITLES | NQF<br>level | Credits | ✓ |
|--------------|--------|--------------|---------|---|
|              |        |              |         |   |

| Activity                     | Evidence of activity will be found<br>where   | Place & planned<br>date of activity | Date<br>Completed |
|------------------------------|-----------------------------------------------|-------------------------------------|-------------------|
| Training                     | Classroom training registers                  | Training Provider<br>Date:          |                   |
| Self assessment              | Assessment contract signed & dated            | Learner file<br>Date:               |                   |
| Assessment<br>contract       | Assessment contract signed & dated            | Learner file<br>Date:               |                   |
| Initial meeting              | Assessor briefing checklist                   | Learner file<br>Date:               |                   |
|                              | Assessment contract & assessment<br>plan      | Learner file<br>Date:               |                   |
| Formative<br>Assessment      | Assessment instruments                        | Learner file<br>Date:               |                   |
| Summative<br>Assessment      | Assessment instruments                        | Learner file<br>Date:               |                   |
| Other Evidence               | Research portfolio (if applicable)            | Learner file<br>Date:               |                   |
| Feedback                     | Feedback Report                               | Learner file<br>Date:               |                   |
| Moderation                   | Moderators report                             | Learner file<br>Date:               |                   |
| Judgement                    | Assessor Summary Report / Moderator<br>report | Learner file<br>Date:               |                   |
| 1 <sup>st</sup> Reassessment | Assessors summary report / instruments        | Learner file<br>Date:               |                   |
| 2 <sup>nd</sup> Reassessment | Assessors summary report / instruments        | Learner file<br>Date:               |                   |

**Special arrangements for assessment**

|           |
|-----------|
| Place     |
| Language  |
| Resources |
| Barriers  |

**People to be involved with assessment**

|                                |                 |
|--------------------------------|-----------------|
| Learner: <u>Xolani Malinga</u> | Manager:        |
| Trainer:                       | Mentor / Coach: |
| Assessor:                      | Moderator:      |

**Next steps for learning**

|  |
|--|
|  |
|  |
|  |

**Resources required for this assessment**

|  |
|--|
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**Guidelines to the learner:**

|                       |
|-----------------------|
| <u>Xolani Malinga</u> |
|                       |
|                       |
|                       |
|                       |
|                       |

Learners Name:

Xolani Colleen Malinga

Learner's signature:



Date: \_\_\_\_\_

Assessors Name: \_\_\_\_\_

Assessor's signature: \_\_\_\_\_  
\_\_\_\_\_

Date: \_\_\_\_\_

### ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
  - If the laid down assessment procedures were not followed during assessments
  - If not all evidence available was taken into account during the assessment
  - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
  - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
  - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

## ASSESSMENT APPEAL APPLICATION FORM

### LEARNER'S NOTICE OF ASSESSMENT APPEAL

**TO: The Internal Moderator**

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name \_\_\_\_\_

Date of submission: \_\_\_\_\_

Name of employee assessed: \_\_\_\_\_

Name of Assessor: \_\_\_\_\_

Date of feedback session: \_\_\_\_\_

### Grounds for Appeal

| No | Tick the applicable ground(s) for appeal                                                                             | Tick |
|----|----------------------------------------------------------------------------------------------------------------------|------|
| 1  | The assessment did not follow the laid down procedure                                                                |      |
| 2  | Not all evidence available was taken into account during the assessment                                              |      |
| 3  | The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process   |      |
| 4  | The assessment was not according to the performance criteria and the range statement stipulated in the unit standard |      |
| 5  | Not all the range items were available for the assessment                                                            |      |

## Reasons for Appeal

| No | Please give detailed reasons for the choice(s) above |
|----|------------------------------------------------------|
| 1  |                                                      |
| 2  |                                                      |
| 3  |                                                      |
| 4  |                                                      |
| 5  |                                                      |

Learner's signature: \_\_\_\_\_

Date: \_\_\_\_\_

\_\_\_\_\_

Employee witness: \_\_\_\_\_

Date: \_\_\_\_\_

\_\_\_\_\_

## PRE-ASSESSMENT MEETING CHECKLIST

**Points Assessor must cover in the initial meeting with the learner - Please tick**

| Item | Points to be covered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Tick                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1    | Welcome the candidate <b>and put them at ease</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <input checked="" type="checkbox"/> |
| 2    | <b>Explain the purpose of the meeting</b> (why you are there and how long the meeting will take)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <input checked="" type="checkbox"/> |
| 3    | <b>Explain the</b> <ul style="list-style-type: none"> <li>▪ NQF</li> <li>▪ Credits</li> <li>▪ Certification process</li> <li>▪ Learning pathways</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         | <input checked="" type="checkbox"/> |
| 4    | <b>Explain</b> <ul style="list-style-type: none"> <li>▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators)</li> <li>▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)</li> </ul>                                                                                                                                                                                                                                                                                                                       | <input checked="" type="checkbox"/> |
| 5    | <b>Explain the assessment process?</b> <ul style="list-style-type: none"> <li>▪ Check learner readiness for assessment (logbook / self assessment)</li> <li>▪ Assessment contract to be completed</li> <li>▪ Preparation of learner (this meeting)</li> <li>▪ The assessment (observation and knowledge questionnaire)</li> <li>▪ Judgement of the evidence</li> <li>▪ Outcome of assessment (competent, not yet competent, need further evidence)</li> </ul>                                                                                                                       | <input checked="" type="checkbox"/> |
| 6    | <b>Give Learner copies of the following documentation and explain each document</b> <ul style="list-style-type: none"> <li>▪ The Assessment Guide which includes                             <ul style="list-style-type: none"> <li>○ The relevant unit standard (s)</li> <li>○ Assessment contract</li> <li>○ Assessment plan</li> <li>○ Observation checklist</li> <li>○ Knowledge checklist</li> </ul> </li> </ul>                                                                                                                                                               | <input checked="" type="checkbox"/> |
| 7    | <b>Discuss the assessment plan</b> (complete the assessment plan document) <ul style="list-style-type: none"> <li>▪ Allow the learner to participate in the decisions made</li> <li>▪ Agree on dates, time and venue for the assessment and feedback</li> <li>▪ Agree on evidence the learner can submit</li> <li>▪ Agree and explain the assessment methods</li> <li>▪ Identify and discuss special assessment needs of the candidate</li> <li>▪ Identify and eliminate unfair barriers (language, disability etc)</li> <li>▪ Discuss and agree on witness requirements</li> </ul> | <input checked="" type="checkbox"/> |
| 8    | <b>Tell the candidate his/her rights and responsibilities, the assessment procedures and policies</b> <ul style="list-style-type: none"> <li>▪ How many times they may be assessed</li> <li>▪ Appeals process / procedure</li> <li>▪ Reassessment policy</li> </ul>                                                                                                                                                                                                                                                                                                                 | <input checked="" type="checkbox"/> |
| 9    | <b>Ensure the assessment environment is appropriate</b> or make special arrangements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <input checked="" type="checkbox"/> |
| 10   | Discuss moderation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <input checked="" type="checkbox"/> |
| 11   | Allow the learner opportunity to clarify any items discussed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <input checked="" type="checkbox"/> |

|                                                                                               |                              |
|-----------------------------------------------------------------------------------------------|------------------------------|
| Learner declaration of acceptance of assessment instruments and relevant documentation: Date: |                              |
| Learners Name: <i>Xolani Colen Malinga</i>                                                    | Signature <i>[Signature]</i> |
| Assessors Name:                                                                               | Signature:                   |

### **INSTRUCTIONS:**

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.



## LEARNING UNIT 1:

### **KM-08-KT01: Low voltage protection (100%)**

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>Learning Outcome 1:</b> | KT0101 Purpose and application of low voltage protection            |
| <b>Learning Outcome 2:</b> | KT0102 Types of low voltage protection                              |
| <b>Learning Outcome 3:</b> | KT0103 Low voltage protection parameters and statutory requirements |

| Task | Questions Description                                         | Marks |
|------|---------------------------------------------------------------|-------|
| 1    | Name and describe the types of low voltage protective devices | 20    |

Fuses: A fuse is a simple protective device that contains a thin metal wire.  
When excessive current flows (overload or short circuit) the wire melts and breaks the circuits.

Circuit breaker (CBs)  
Automatically switches off the electrical circuit when a fault is detected.  
Protects against overloads and short circuits.

Residual Current Devices (RCD) / earth leakage circuit breakers (ELCB)  
Detects leakage current flowing to earth.  
Protects people from electric shock and prevents fire hazards.

Overload Relays

Used mainly with motors.  
Protects motors from overheating due to overload.

Surge Protection devices (SPDs)

Protect against voltage spikes (surges).  
Protects sensitive electronic equipment.

## Isolators (Disconnect Switches)

Used to manually disconnect a circuit from power supply.  
Safety during servicing and repairs.

## Contactors with Protection.

Electrically controlled switches used in motor control circuits.  
Controls and protects electrical circuits, especially motors.

Low-Voltage protective devices are essential for

Preventing damage to equipment

Protecting people from electric shock

Avoiding fires caused by electrical faults.

10

| Task | Questions Description                                                                              | Marks |
|------|----------------------------------------------------------------------------------------------------|-------|
| 2    | Describe the effect of adverse conditions on the operational characteristics of protective devices | 10    |

High temperature.

Causes overheating of Components.

Low Temperature

Makes materials stiff or brittle

Mechanical parts may operate slowly or fail to trip when required.

Moisture / Humidity.

Leads to corrosion and insulation failure

May cause leakage currents, false tripping of RCDs or short circuits

Dust and dirt

Accumulates inside devices.

Can block moving parts, cause overheating.

(8)

Vibration and Mechanical Shock

Common in industrial environments.

Can loosen connections or damage internal components

Overvoltage / Voltage Surges

Caused by lightning or switching

Can damage protective devices or reduce their lifespan.

Ageing and Wear

Components degrade over time.

(1)

Corrosive Environments (chemicals, Salt Air, Air)

Causes rust and deterioration.

Ageing and Wear (can components degrade overtime (fuse elements weaken).

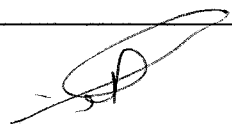
141 Page  
a. Chemical can damage insulation and metal parts.

Leads to Malfunction or complete failure.

## Assessor's Report

|                               |                        |          |                |
|-------------------------------|------------------------|----------|----------------|
| Candidate's Name              | Xolani Collen Malinga  | ID No.   | 9511155424085. |
| Assessor's Name               |                        | Reg. No. |                |
| Unit Standard Title           |                        |          |                |
| ASSESSMENT DECISION           |                        |          |                |
| Outcome                       | C                      | NYC      | Comments       |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
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|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
|                               |                        |          |                |
| OVERALL COMPETENCE JUDGEMENT: |                        |          |                |
| Comments                      |                        |          |                |
| Date                          |                        |          |                |
|                               |                        |          |                |
| Signature of Assessor         | Signature of Candidate |          |                |

## Candidate Feedback Report

|                             |                         |          |                                                                                      |               |
|-----------------------------|-------------------------|----------|--------------------------------------------------------------------------------------|---------------|
| Candidate's Name            | Xolani Collen Malinga   |          | ID No.                                                                               | 9511155424085 |
| Assessor's Name             |                         | Reg. No. |                                                                                      |               |
| Title                       | low Voltage Protection. |          |                                                                                      |               |
| <b>ASSESSMENT DECISION</b>  |                         |          |                                                                                      |               |
| Source of Evidence          | C                       | NYC      | Comment                                                                              |               |
| Knowledge questions         | ✓                       |          | Good                                                                                 |               |
| Product /Work sample        | ✓                       |          | Good                                                                                 |               |
| Project                     | ✓                       |          | Good                                                                                 |               |
| Indirect Evidence           | ✓                       |          | Good                                                                                 |               |
| Overall Assessment Decision |                         |          |                                                                                      |               |
| Additional Notes            |                         |          |                                                                                      |               |
| Date                        |                         |          |                                                                                      |               |
|                             |                         |          |  |               |
| Signature of Assessor       |                         |          | Signature of Candidate                                                               |               |

# Candidate Appeal Form

|                                                                 |  |          |  |
|-----------------------------------------------------------------|--|----------|--|
| Candidate's Name:                                               |  | ID No.   |  |
| Assessor's Name:                                                |  | Reg. No. |  |
| Title:                                                          |  |          |  |
| Date:                                                           |  |          |  |
| <b>SECTION 1</b>                                                |  |          |  |
| Candidate's reason for disagreeing with the assessment decision |  |          |  |
| Assessor's rationale for the assessment decision                |  |          |  |
| Candidate's signature                                           |  |          |  |
| Assessor's signature                                            |  |          |  |
| <b>SECTION 2</b>                                                |  |          |  |
| Moderator's decision                                            |  |          |  |
| Moderator's signature                                           |  |          |  |

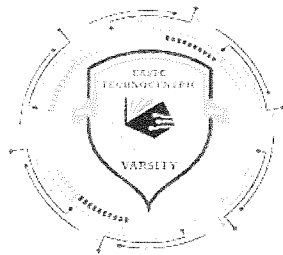
## Moderator's Report

|                             |                                                                                     |          |                     |
|-----------------------------|-------------------------------------------------------------------------------------|----------|---------------------|
| Moderator's Name            |                                                                                     | Reg. No. |                     |
| Assessor's Name             |                                                                                     | Reg. No. |                     |
| Candidate's Name            | <i>Robeni Colleen Malinga</i>                                                       | ID No.   | <i>951155424085</i> |
| Title                       |                                                                                     |          |                     |
| <b>MODERATION DECISION</b>  |                                                                                     |          |                     |
| Outcome                     | C                                                                                   | NYC      | Comments            |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
|                             |                                                                                     |          |                     |
| Overall Moderation Decision |                                                                                     |          |                     |
| Feedback to Assessor        |                                                                                     |          |                     |
| Action Required             |                                                                                     |          |                     |
| Date of Moderation          |                                                                                     |          |                     |
| Signature of Moderator      |                                                                                     |          |                     |
| Signature of Assessor       |                                                                                     |          |                     |
| Signature of Candidate      |  |          |                     |

## ASSESSMENT REVIEW

| NAME of LEARNER <i>Xolani Collen Malinga</i>                                                                         |                                                                     | NAME of ASSESSOR                                                               |             |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------|
| VENUE <i>East Technogetic University</i>                                                                             |                                                                     | DATE of REVIEW                                                                 |             |
|                                                                                                                      |                                                                     |                                                                                |             |
| Review Dimension                                                                                                     | ASSESSOR                                                            | LEARNER/<br>CANDIDATE                                                          | ACTION      |
| The principles/criteria for good assessment were achieved?                                                           | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The assessment related to the registered unit standard?                                                              | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The assessment was practical?                                                                                        | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| It was time efficient and cost-effective and did not interfere with my normal responsibilities?                      | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The assessment instruments were fair, clear and understandable                                                       | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The assessment judgements was made against set requirements                                                          | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The venue and equipment was functional?                                                                              | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| Special needs were identified and the assessment plan was adjusted                                                   | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| Feedback was constructive against the evidence required                                                              | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| An opportunity to appeal was given                                                                                   | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| The evidence was recorded                                                                                            | <input type="checkbox"/> Agree<br><input type="checkbox"/> Disagree | <input checked="" type="checkbox"/> Agree<br><input type="checkbox"/> Disagree |             |
| <b>LEARNER'S DECLARATION OF UNDERSTANDING</b>                                                                        |                                                                     |                                                                                |             |
| I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid |                                                                     |                                                                                |             |
|                                                                                                                      |                                                                     |                                                                                |             |
| <b>Learner</b>                                                                                                       | <b>Date</b>                                                         | <b>Assessor</b>                                                                | <b>Date</b> |
| <b>Moderator</b>                                                                                                     |                                                                     | <b>Date</b>                                                                    |             |





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|                                     |                                                  |
|-------------------------------------|--------------------------------------------------|
| <b>Module #</b>                     | 671101000-KM-08:                                 |
| <b>NQF Level</b>                    | level 4,                                         |
| <b>Notional hours</b>               | 50                                               |
| <b>Credit(s)</b>                    | Cr 5                                             |
| <b>Occupation or Specialisation</b> | 671101000: Occupational Certificate: Electrician |
| <b>Associated Occupation</b>        | 671101: Electrician                              |

## LOW VOLTAGE PROTECTION

### FORMATIVE ASSESSMENT

| Details         | Please Complete this Section |
|-----------------|------------------------------|
| Name            | Xolani Collen Malinga        |
| Contact Address | 32467 Mandela Str            |
| Telephone (H)   | 0747732317                   |
| Telephone (W)   | 0747732317                   |
| Facsimile       | N/A                          |
| Cellular        | 0747732317                   |
| E-mail          | collenx14@gmail.com          |

### About the Formative Assessment

|                                     |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>                      | The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.                                                                                                                   |
| <b>Context</b>                      | This assessment represents the Formative Assessment component and should be completed in the classroom/training room.                                                                                                                                                                                                                                               |
| <b>Resources</b>                    | The following are resources needed for this assessment: <ol style="list-style-type: none"> <li>1. Learner Guide; and</li> <li>2. Assessment Preparation.</li> </ol>                                                                                                                                                                                                 |
| <b>Instructions to Facilitators</b> | Facilitators will be required to: <ul style="list-style-type: none"> <li>• Explain the completion of the Formative Assessment to each learner; and</li> <li>• Interview the learner on similar questions, should he/she not be able to write.</li> </ul>                                                                                                            |
| <b>Instructions to Learners</b>     | Learners will be required to: <ul style="list-style-type: none"> <li>• Complete the Formative Assessment as per the instructions;</li> <li>• Ensure that all questions are completed;</li> <li>• Ensure that the completion of the workbook is their own work;</li> <li>• Ensure that all annexure are attached to the workbook and clearly referred to;</li> </ul> |
| <b>Assessment Time</b>              | Learners are required to complete this assessment within the allocated time frame/hours.                                                                                                                                                                                                                                                                            |

|                   |                                                                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Mark</b> | This formative assessment carries a total mark of _ <b>points</b> . In order to meet the pass mark, learners are required to achieve a minimum of <b>80%</b> of the total marks. |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                |
|--------------------------------|
| <b>LEARNER ASSESSMENT PLAN</b> |
|--------------------------------|

| Unit code | /MODULE TITLES | NQF level | Credits | ✓ |
|-----------|----------------|-----------|---------|---|
|           |                |           |         |   |

| Activity                     | Evidence of activity will be found where   | Place & planned date of activity | Date Completed |
|------------------------------|--------------------------------------------|----------------------------------|----------------|
| Training                     | Classroom training registers               | Training Provider<br>Date:       |                |
| Self assessment              | Assessment contract signed & dated         | Learner file<br>Date:            |                |
| Assessment contract          | Assessment contract signed & dated         | Learner file<br>Date:            |                |
| Initial meeting              | Assessor briefing checklist                | Learner file<br>Date:            |                |
| Unit Standard No             | Assessment contract & assessment plan      | Learner file<br>Date:            |                |
| Formative Assessment         | Assessment instruments                     | Learner file<br>Date:            |                |
| Summative Assessment         | Assessment instruments                     | Learner file<br>Date:            |                |
| Other Evidence               | Research portfolio (if applicable)         | Learner file<br>Date:            |                |
| Feedback                     | Feedback Report                            | Learner file<br>Date:            |                |
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| 2 <sup>nd</sup> Reassessment | Assessors summary report / instruments     | Learner file<br>Date:            |                |

**Special arrangements for assessment**

|           |
|-----------|
| Place     |
| Language  |
| Resources |
| Barriers  |

**People to be involved with assessment**

|                                |                 |
|--------------------------------|-----------------|
| Learner: <u>Xolani Malinga</u> | Manager:        |
| Trainer:                       | Mentor / Coach: |
| Facilitator                    |                 |
| Assessor:                      | Moderator:      |

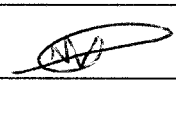
**Next steps for learning**

|  |
|--|
|  |
|  |
|  |

**Resources required for this assessment**

|  |
|--|
|  |
|  |
|  |
|  |
|  |

**Guidelines to the learner:**

|                                                                                                            |
|------------------------------------------------------------------------------------------------------------|
| <u>Xolani Malinga</u>  |
|                                                                                                            |
|                                                                                                            |
|                                                                                                            |
|                                                                                                            |

Learners Name: Xolani Collen MalingaLearner's signature: 

Date: \_\_\_\_\_

Facilitator Name: \_\_\_\_\_

Facilitator's signature: \_\_\_\_\_

Date: \_\_\_\_\_

### ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
  - If the laid down assessment procedures were not followed during assessments
  - If not all evidence available was taken into account during the assessment
  - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
  - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
  - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

## ASSESSMENT APPEAL APPLICATION FORM

### LEARNER'S NOTICE OF ASSESSMENT APPEAL

**TO: The Internal Moderator**

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name \_\_\_\_\_

Date of submission: \_\_\_\_\_

Name of employee assessed: \_\_\_\_\_

Name of Assessor: \_\_\_\_\_

Date of feedback session: \_\_\_\_\_

### Grounds for Appeal

| No | Tick the applicable ground(s) for appeal                                                                             | Tick |
|----|----------------------------------------------------------------------------------------------------------------------|------|
| 1  | The assessment did not follow the laid down procedure                                                                |      |
| 2  | Not all evidence available was taken into account during the assessment                                              |      |
| 3  | The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process   |      |
| 4  | The assessment was not according to the performance criteria and the range statement stipulated in the unit standard |      |
| 5  | Not all the range items were available for the assessment                                                            |      |

### Reasons for Appeal

| No | Please give detailed reasons for the choice(s) above |
|----|------------------------------------------------------|
| 1  |                                                      |
| 2  |                                                      |
| 3  |                                                      |
| 4  |                                                      |
| 5  |                                                      |

|  |  |
|--|--|
|  |  |
|--|--|

Learner's signature: \_\_\_\_\_

Date: \_\_\_\_\_

Employee witness: \_\_\_\_\_

Date: \_\_\_\_\_

### PRE-ASSESSMENT MEETING CHECKLIST

**Points Assessor must cover in the initial meeting with the learner - Please tick**

| Item | Points to be covered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Tick                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1    | Welcome the candidate <b>and put them at ease</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <input checked="" type="checkbox"/> |
| 2    | <b>Explain the purpose of the meeting</b> (why you are there and how long the meeting will take)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <input checked="" type="checkbox"/> |
| 3    | <b>Explain the</b> <ul style="list-style-type: none"> <li>▪ NQF</li> <li>▪ Credits</li> <li>▪ Certification process</li> <li>▪ Learning pathways</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         | <input checked="" type="checkbox"/> |
| 4    | <b>Explain</b> <ul style="list-style-type: none"> <li>▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators)</li> <li>▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)</li> </ul>                                                                                                                                                                                                                                                                                                                       | <input checked="" type="checkbox"/> |
| 5    | <b>Explain the assessment process?</b> <ul style="list-style-type: none"> <li>▪ Check learner readiness for assessment (logbook / self assessment)</li> <li>▪ Assessment contract to be completed</li> <li>▪ Preparation of learner (this meeting)</li> <li>▪ The assessment (observation and knowledge questionnaire)</li> <li>▪ Judgement of the evidence</li> <li>▪ Outcome of assessment (competent, not yet competent, need further evidence)</li> </ul>                                                                                                                       | <input checked="" type="checkbox"/> |
| 6    | <b>Give Learner copies of the following documentation and explain each document</b> <ul style="list-style-type: none"> <li>▪ The Assessment Guide which includes               <ul style="list-style-type: none"> <li>○ The relevant unit standard (s)</li> <li>○ Assessment contract</li> <li>○ Assessment plan</li> <li>○ Observation checklist</li> <li>○ Knowledge checklist</li> </ul> </li> </ul>                                                                                                                                                                             | <input checked="" type="checkbox"/> |
| 7    | <b>Discuss the assessment plan</b> (complete the assessment plan document) <ul style="list-style-type: none"> <li>▪ Allow the learner to participate in the decisions made</li> <li>▪ Agree on dates, time and venue for the assessment and feedback</li> <li>▪ Agree on evidence the learner can submit</li> <li>▪ Agree and explain the assessment methods</li> <li>▪ Identify and discuss special assessment needs of the candidate</li> <li>▪ Identify and eliminate unfair barriers (language, disability etc)</li> <li>▪ Discuss and agree on witness requirements</li> </ul> | <input checked="" type="checkbox"/> |
| 8    | <b>Tell the candidate his/her rights and responsibilities, the assessment procedures and policies</b> <ul style="list-style-type: none"> <li>▪ How many times they may be assessed</li> <li>▪ Appeals process / procedure</li> <li>▪ Reassessment policy</li> </ul>                                                                                                                                                                                                                                                                                                                 | <input checked="" type="checkbox"/> |

| Item | Points to be covered                                                          | Tick |
|------|-------------------------------------------------------------------------------|------|
| 9    | Ensure the assessment environment is appropriate or make special arrangements | ✓    |
| 10   | Discuss moderation                                                            | ✓    |
| 11   | Allow the learner opportunity to clarify any items discussed                  | ✓    |

|                                                                                               |                                                                                               |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Learner declaration of acceptance of assessment instruments and relevant documentation: Date: |                                                                                               |
| Learners Name: Yolani Malinga                                                                 | Signature:  |
| Facilitator Name:                                                                             | Signature:                                                                                    |

Complete the following questions as per instructions provided

## **LEARNING UNIT 1:**

### **KM-08-KT01: Low voltage protection (100%)**

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>Learning Outcome 1:</b> | KT0101 Purpose and application of low voltage protection            |
| <b>Learning Outcome 2:</b> | KT0102 Types of low voltage protection                              |
| <b>Learning Outcome 3:</b> | KT0103 Low voltage protection parameters and statutory requirements |

| <b>Task</b> | <b>Questions Description</b>                                                                                                                                                       | <b>Marks</b> |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>1.</b>   | Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. | <b>10</b>    |

Overload Relays.

Function: protect electrical motors and circuits from overcurrent due to overload (When current exceeds the normal rated value over time)

Operation: Works on thermal or electronic principle

In the thermal overload relays, a bimetal strip bends when heated by excess current

this bending opens the control circuit stopping the motor. It does not trip instantly, allowing temporary overloads

Fuse

Function: Protect circuits from excess current and short circuit.

Operation: Contains a thin metal wire or strip

When current exceeds the rated value, the wire

10

heats up and melts, breaking the circuit  
Once blown, the fuse must be replaced  
Provides fast and reliable protection

### Circuit breakers

Function: Protect circuits from overload and short circuit faults and can also be used as a switch

Operation: Uses electromagnetic and thermal mechanisms  
During overload  $\rightarrow$  thermal element trips slowly  
During short circuit  $\rightarrow$  magnetic mechanism trips instantly  
Can be reset after tripping, unlike a fuse

### Earth leakage Protection devices (ELPD) (RCD)

Function: Protect people from electric shock and prevent fire by detecting leakage current to earth.

Operation: Monitors the current flowing in live and neutral conductors

If there is a difference (leakage) it trips the circuit  
Very sensitive (e.g. 30mA for humans protection)

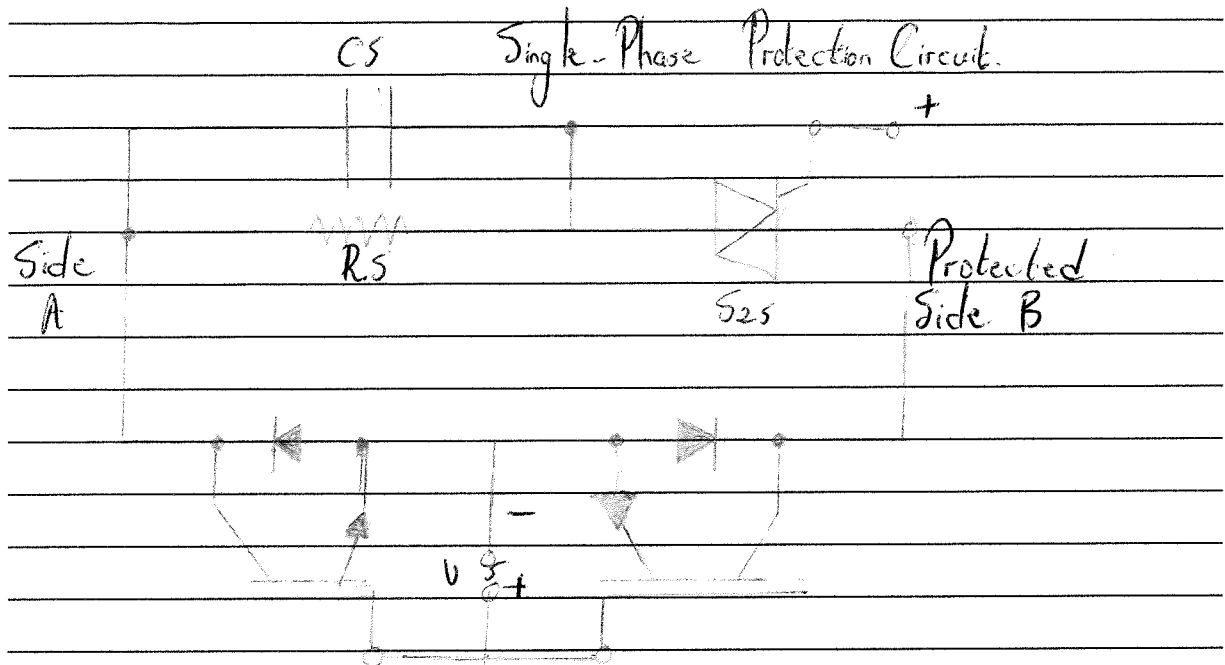
Overload relays protect against overload (mainly motors).

Fuse: protect against overcurrent by melting.

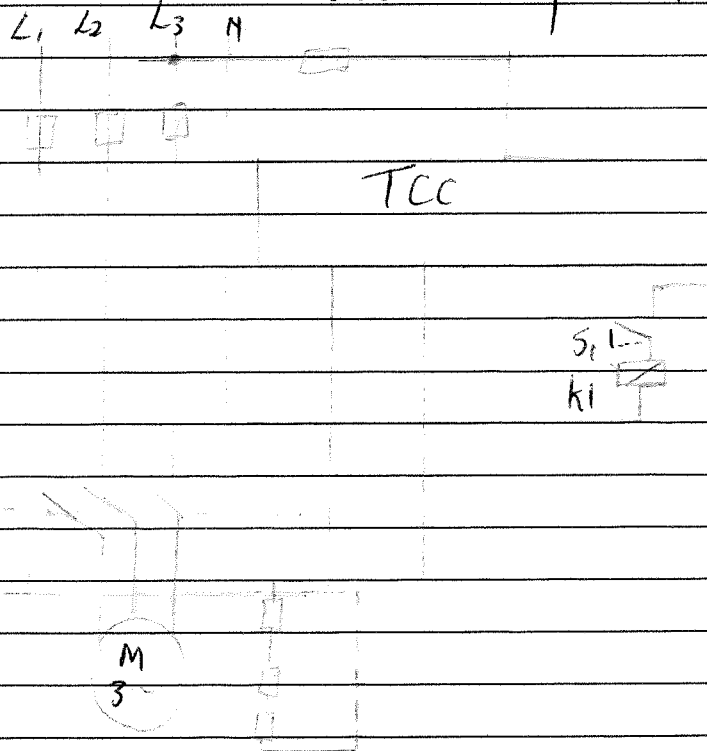
Circuit breakers protect and can be reset.

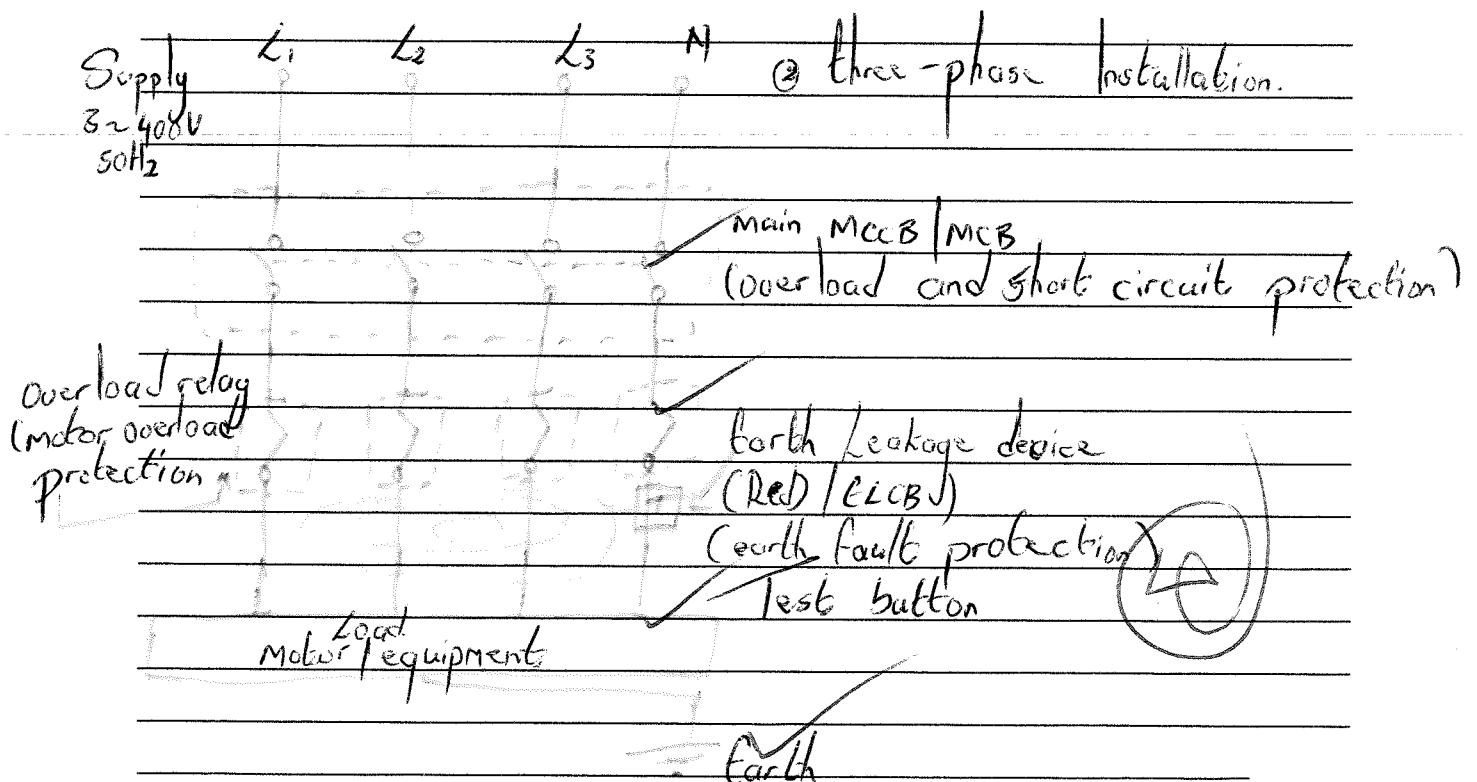
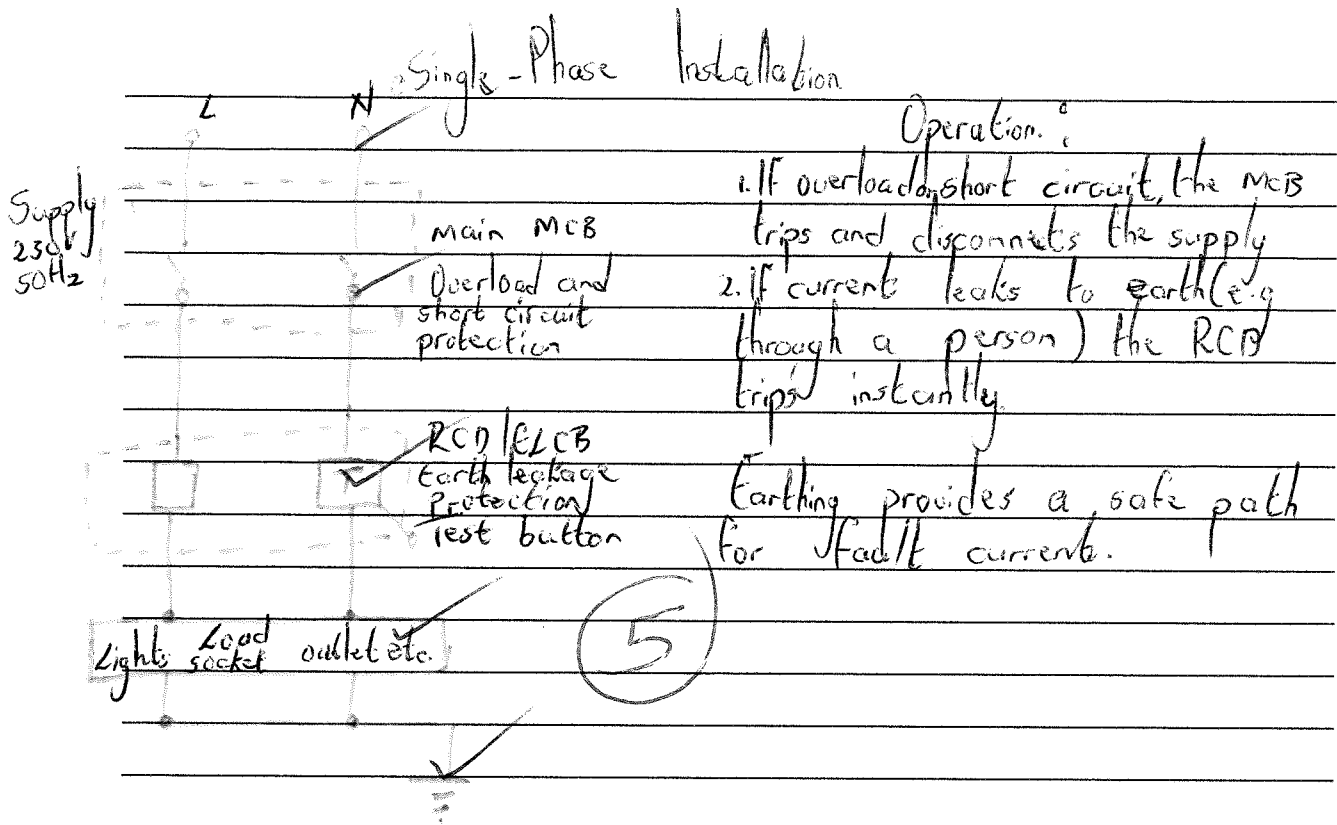
Earth leakage devices protect people from shock and leakage faults.

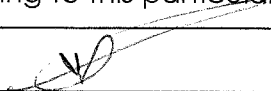
| Task | Questions Description                                                                                         | Marks |
|------|---------------------------------------------------------------------------------------------------------------|-------|
| 2    | Explain, with the aid of circuit diagrams, how single-and three phase electrical installations are protected. | 10    |



Electrical Protection of 3phase Motor





| DECLARATION BY LEARNER                                                                                                                                                                                                                                                               |             |                    |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|-------------|
| I, <u>Idani Collen Malinga</u> declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event. |             |                    |             |
|                                                                                                                                                                                                     |             |                    |             |
| <b>Learner</b>                                                                                                                                                                                                                                                                       | <b>Date</b> | <b>Facilitator</b> | <b>Date</b> |

| DECLARATION BY THE ASSESSOR                                                                                                                      |             |                  |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|-------------|
| I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment. |             |                  |             |
|                                                                                                                                                  |             |                  |             |
| <b>Assessor</b>                                                                                                                                  | <b>Date</b> | <b>Moderator</b> | <b>Date</b> |

### **Application for Assessment**

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: \_\_\_\_\_

Contact Number: \_\_\_\_\_

Interpreter required: Yes/No

Name: \_\_\_\_\_

Contact Number: \_\_\_\_\_

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

### **Recommendation for Assessment**

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

### **Notification of Assessment**

Assessment will take place as follows:

|             | <b>Theory</b>      | <b>Practical</b>     |
|-------------|--------------------|----------------------|
| Date        |                    |                      |
| Time        | 9H00               | 9H00                 |
| Venue       | Employer premises  | Employer premises    |
| Instruments | Written assignment | Practical Assessment |

#### Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

### Assessment Planning

|                                                 | <u>Theory</u> | <u>Practical</u> |
|-------------------------------------------------|---------------|------------------|
| Date of assessment:                             |               |                  |
| When would be the best time for the Assessment? | 9H00          | 9H00             |

|                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Have you received an application to assess the candidate? Yes                                                                                                                                                                                                                                                                                                                                 |
| What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.<br><br>1. Written Assignment : Printed Questionnaire<br><br>2. Observation : As per Practical Assessment Guide<br><br>3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation                                                             |
| What resources would you require to conduct the assessment?<br><br>1. Assessor toolkit<br><br>2. Theoretical assignment<br><br>3. Practical assessment<br><br>4. Learner toolkit                                                                                                                                                                                                              |
| How would you sequence your activities? Explain<br><br>1. Record preparation process<br><br>2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment<br><br>3. Learner will write summative assessments<br><br>4. Conduct observation in workplace where applicable<br><br>5. Complete learner toolkit<br><br>6. Complete assessor toolkit |



